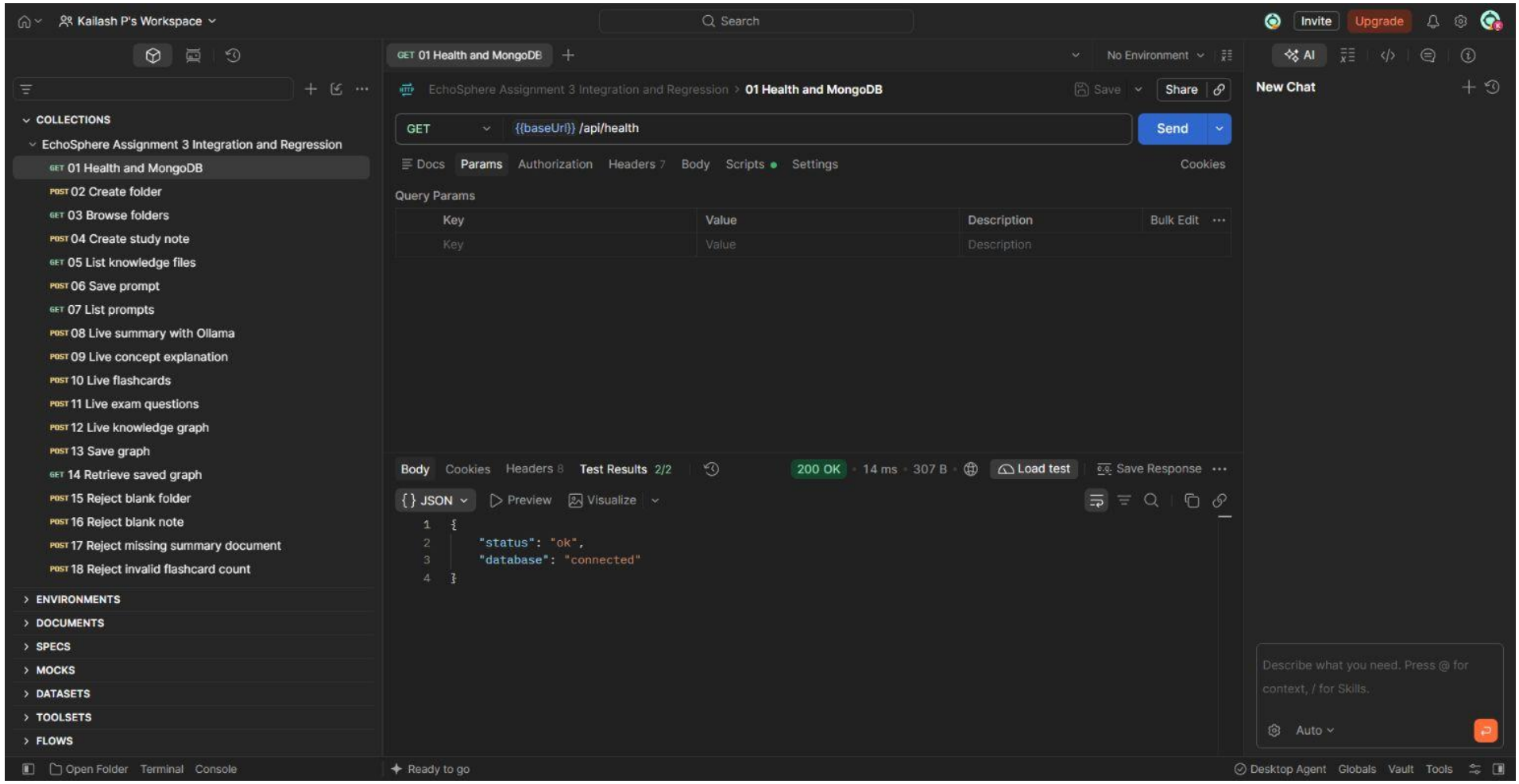
EchoSphere Assignment 3 Testing Evidence

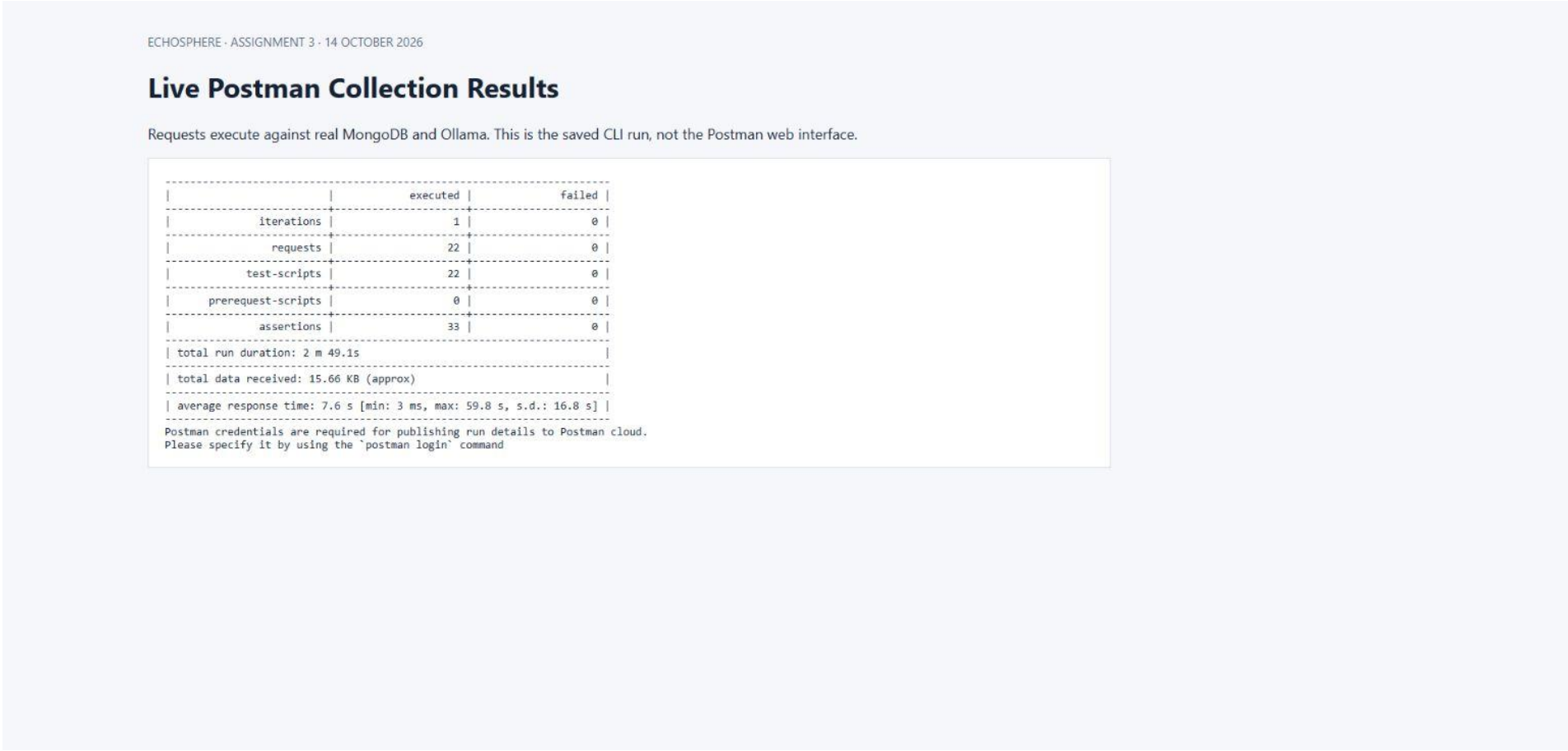
Testing and build verification completed on 14 October 2026 using disposable assignment data. The following screenshots record the observed results.

Integration Testing



Postman returns HTTP 200 and passes both health assertions, confirming that Express is connected to MongoDB.

Integration Testing with Live AI



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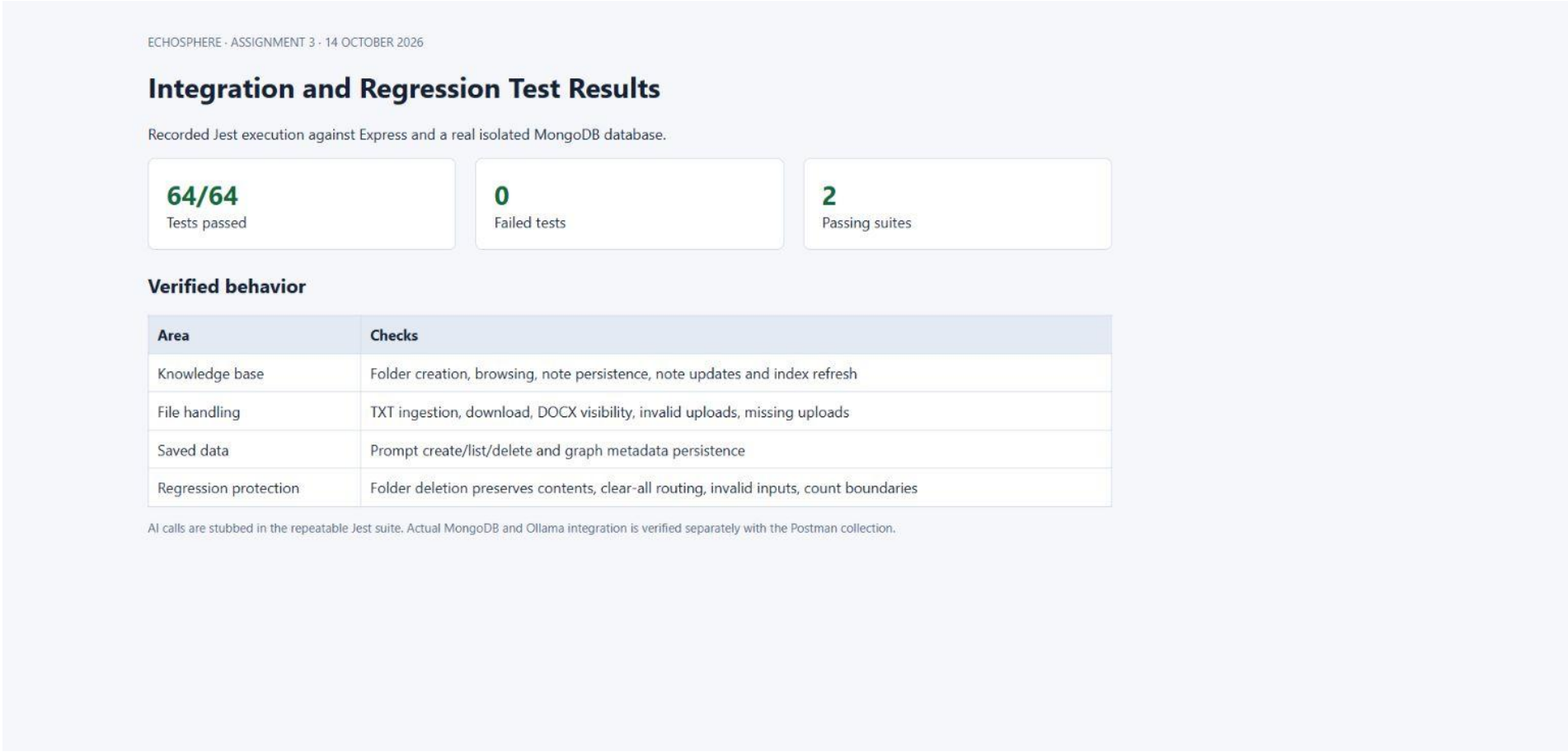
Live Postman Collection Results

Requests execute against real MongoDB and Ollama. This is the saved CLI run, not the Postman web interface.

```
┌─────────────────────────┬────────────────────┬───────────────────┐
│                         │           executed │            failed │
├─────────────────────────┼────────────────────┼───────────────────┤
│              iterations │                  1 │                 0 │
├─────────────────────────┼────────────────────┼───────────────────┤
│                requests │                 22 │                 0 │
├─────────────────────────┼────────────────────┼───────────────────┤
│            test-scripts │                 22 │                 0 │
├─────────────────────────┼────────────────────┼───────────────────┤
│      prerequest-scripts │                  0 │                 0 │
├─────────────────────────┼────────────────────┼───────────────────┤
│              assertions │                 33 │                 0 │
├─────────────────────────┴────────────────────┴───────────────────┤
│ total run duration: 2 m 49.1s                                    │
├──────────────────────────────────────────────────────────────────┤
│ total data received: 15.66 KB (approx)                           │
├──────────────────────────────────────────────────────────────────┤
│ average response time: 7.6 s [min: 3 ms, max: 59.8 s, s.d.: 16.8 s] │
└──────────────────────────────────────────────────────────────────┘
Postman credentials are required for publishing run details to Postman cloud.
Please specify it by using the `postman login` command
```

The saved Postman CLI execution passes all 33 assertions across 22 requests against real MongoDB and Ollama.

Regression Testing



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Integration and Regression Test Results

Recorded Jest execution against Express and a real isolated MongoDB database.

64/64	0	2
Tests passed	Failed tests	Passing suites

Verified behavior

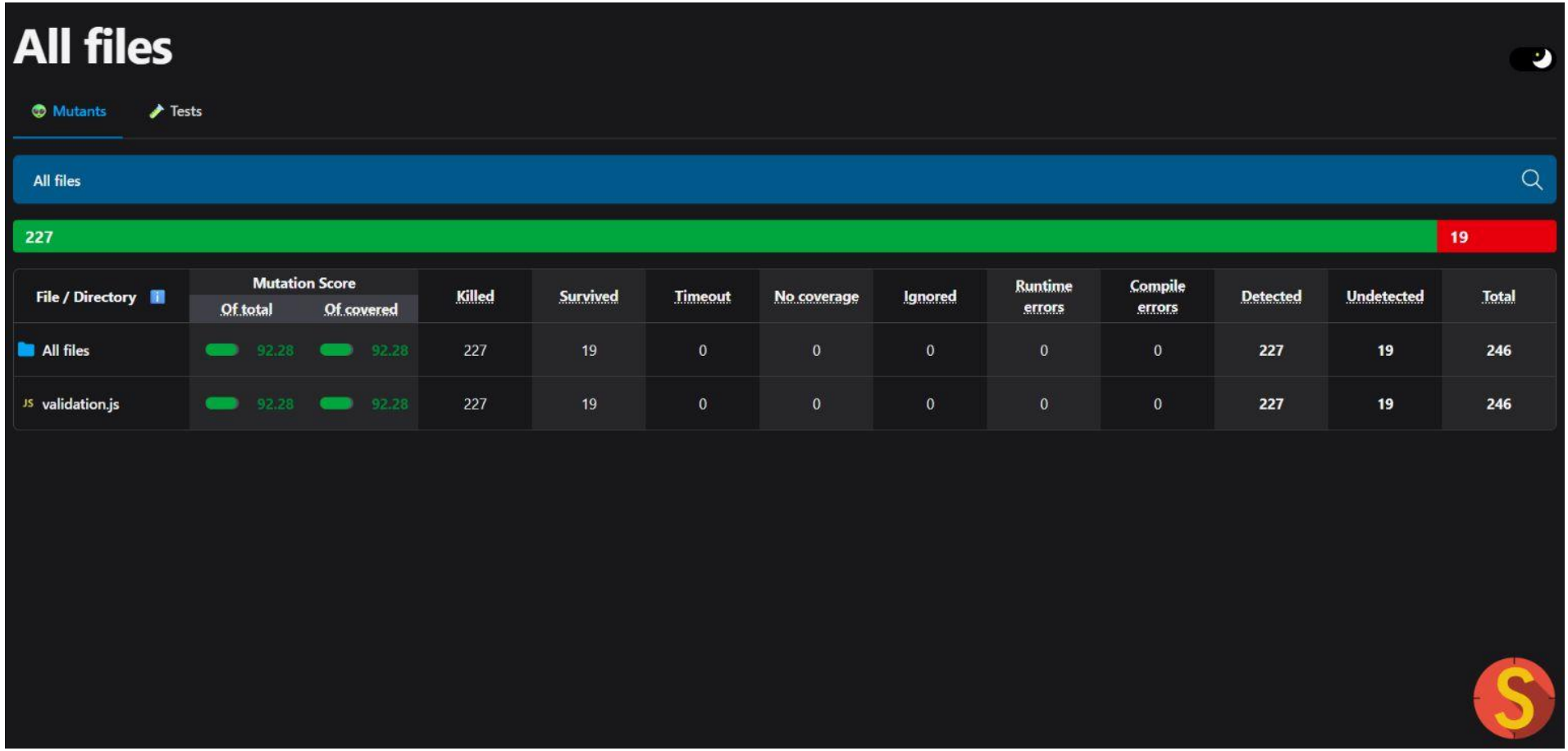
Area	Checks
Knowledge base	Folder creation, browsing, note persistence, note updates and index refresh
File handling	TXT ingestion, download, DOCX visibility, invalid uploads, missing uploads
Saved data	Prompt create/list/delete and graph metadata persistence
Regression protection	Folder deletion preserves contents, clear-all routing, invalid inputs, count boundaries

AI calls are stubbed in the repeatable Jest suite. Actual MongoDB and Ollama integration is verified separately with the Postman collection.

Jest passes 64 tests covering persistence, file handling, input validation and previously broken workflows.

The repeatable Jest integration suite uses real MongoDB with AI calls stubbed; the separate live Postman run exercises the actual AI services.

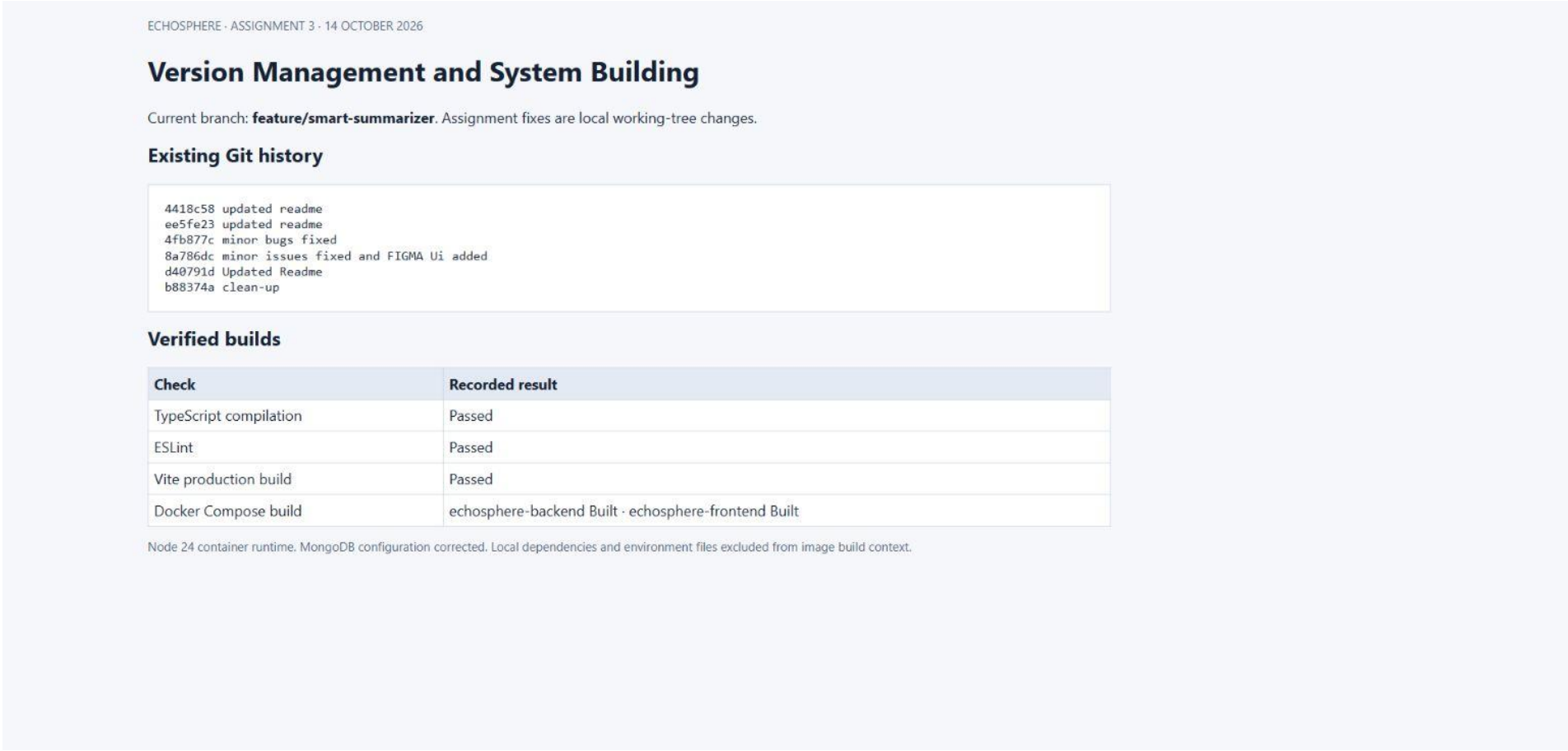
Mutation Testing



Stryker detects 227 of 246 mutations in backend validation with a 92.28 percent mutation score.

Scope is the API input-validation module. Nineteen mutations survive; this is not a project-wide mutation score.

Version Management and System Building



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Version Management and System Building

Current branch: **feature/smart-summarizer**. Assignment fixes are local working-tree changes.

Existing Git history

```
4418c58 updated readme
ee5fe23 updated readme
4fb877c minor bugs fixed
8a786dc minor issues fixed and FIGMA Ui added
d40791d Updated Readme
b88374a clean-up
```

Verified builds

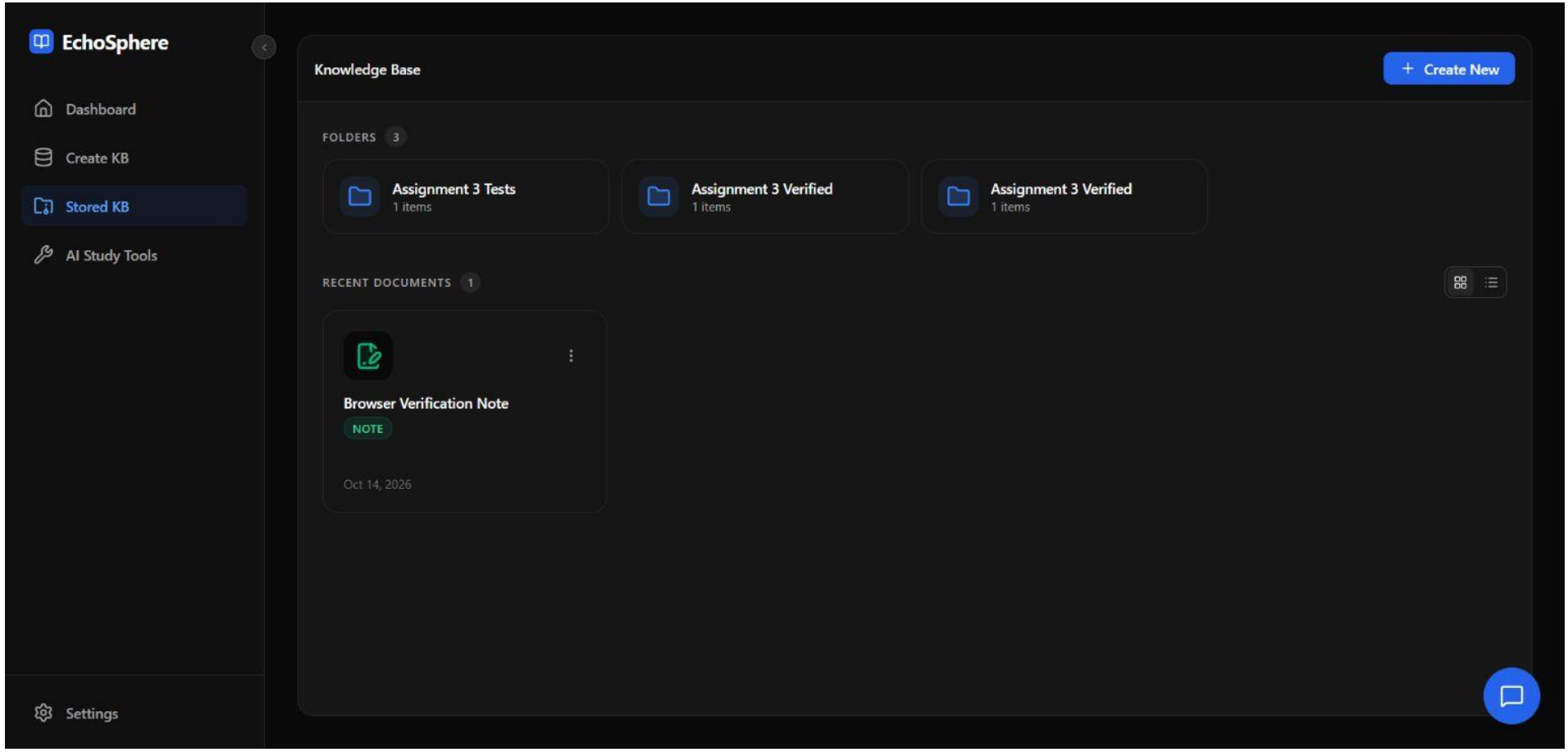
Check	Recorded result
TypeScript compilation	Passed
ESLint	Passed
Vite production build	Passed
Docker Compose build	echosphere-backend Built · echosphere-frontend Built

Node 24 container runtime. MongoDB configuration corrected. Local dependencies and environment files excluded from image build context.

Git history is recorded and TypeScript, ESLint, Vite and both Docker image builds pass.

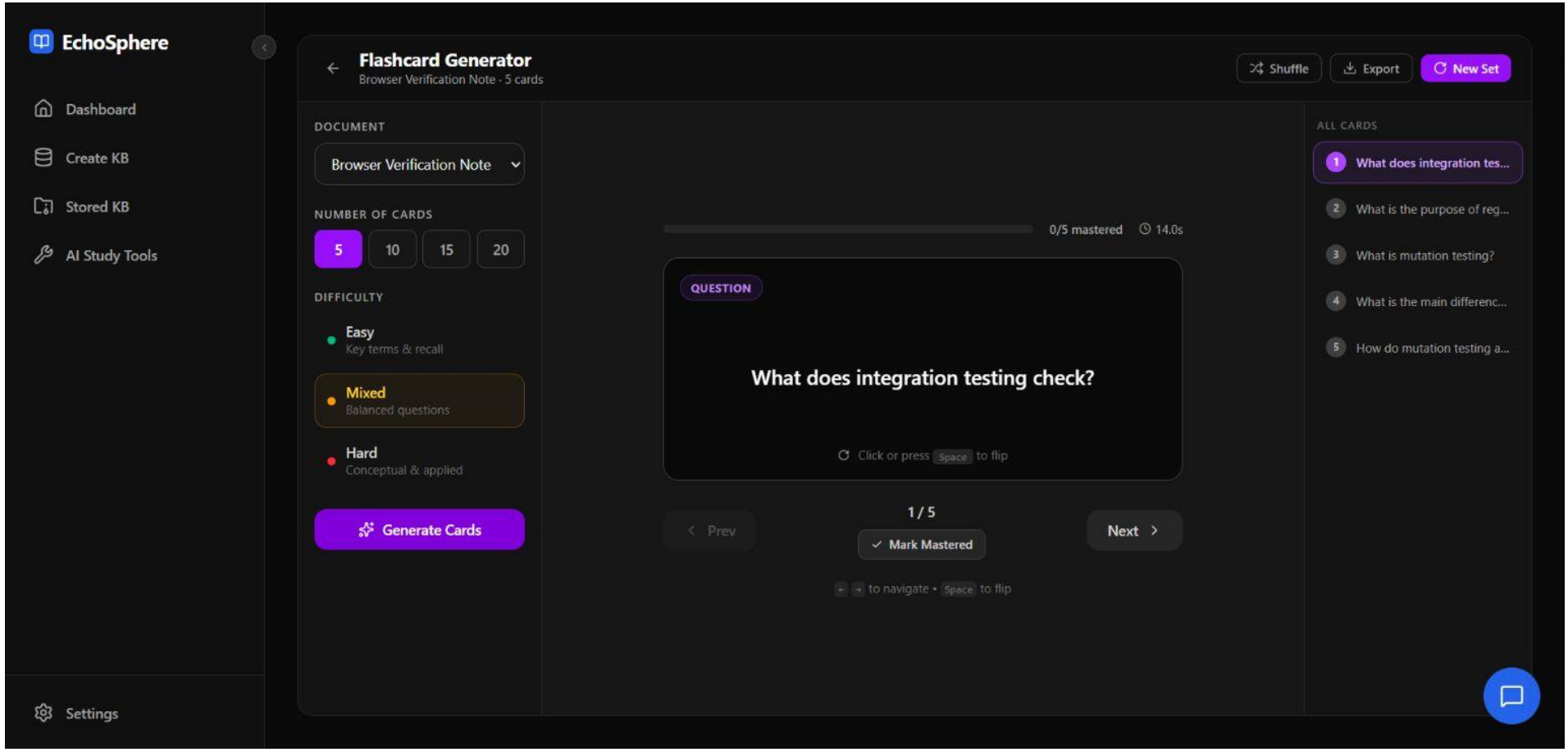
The assignment changes remain local on the existing feature branch and have not been committed or pushed.

Developed Functionality Knowledge Base



A note created through the browser appears in the stored knowledge base after the API saves it to MongoDB.

Developed Functionality Flashcards



Ollama generates study flashcards from the selected note and the browser displays the resulting questions.

Tools and Technologies Used

Application React 19, TypeScript, Vite, Tailwind CSS, Express 5 and Node.js 24.

Data and AI MongoDB with Mongoose, LanceDB vector search, Ollama llama3.2:1b and nomic-embed-text.

Testing Postman CLI and Postman web interface with Desktop Agent, Jest, Supertest and Stryker Mutator.

Build and Evidence Git, Docker Compose, browser screenshots and an editable Word evidence document.

Test Coverage

Integration: health, folders, notes, upload and download, saved prompts, graph persistence and live AI tools.

Regression: TXT ingestion, DOCX file visibility, note reindexing, folder-content preservation, clear-all routing and malformed inputs.

Mutation: required text fields, identifier formats, conversation history, graph structure and count boundaries.

Browser: create a note, verify stored content, generate flashcards and display the results.

Reproducing the Tests

Run npm test in backend with the isolated MongoDB test container available on port 27018. Run npm run test:mutation for the focused mutation analysis. Import EchoSphere.postman_collection.json and run it against the assignment backend on port 5000 with Ollama running.